Agentic AI Assistance for Content Creators

END SEMESTER PROJECT REVIEW

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INTRODUCTION

- In recent years, Professionals across domains are increasingly expected to create digital content for communication, education, and outreach.
- However, **Content Creation is not a single task**; it involves:
  - Information retrieval (Research)
  - Script formulation
  - Recording
  - Editing
  - Publishing
- This Project proposes an **Agentic AI–based system** that assists professionals by **automating research and script generation, video generation , editing, publishing in social media, monitoring logs**, while still keeping humans in control through **approval loops**.

MOTIVATION

- The motivation for this project arises from real-world challenges faced by professionals :
  - Severe **time constraints**
  - Cognitive overload due to repetitive Research tasks, Recording, Editing, Publishing.
  - Inconsistent content quality
  - Lack of integrated AI-assisted tools
- Key motivating factors:
  - Time is the **critical differentiator** in modern professional workflows
  - AI should **assist decision-making**, not replace humans
  - Existing tools are fragmented and non-cohesive
  - Need for **agent-based systems** that think, retrieve, generate, and wait for approval

LITERATURE SURVEY

Existing Approaches Studied

Rule-Based Automation Systems



- Perform predefined tasks
- No intelligence or adaptability

Standalone AI Chatbots



- Generate text
- Lack real-time data grounding
- No workflow orchestration

Traditional Content Management Systems



- Manual data input
- No AI-assisted research or generation

LLM-Based Generators (Without RAG)



- Prone to hallucinations
- Limited domain reliability

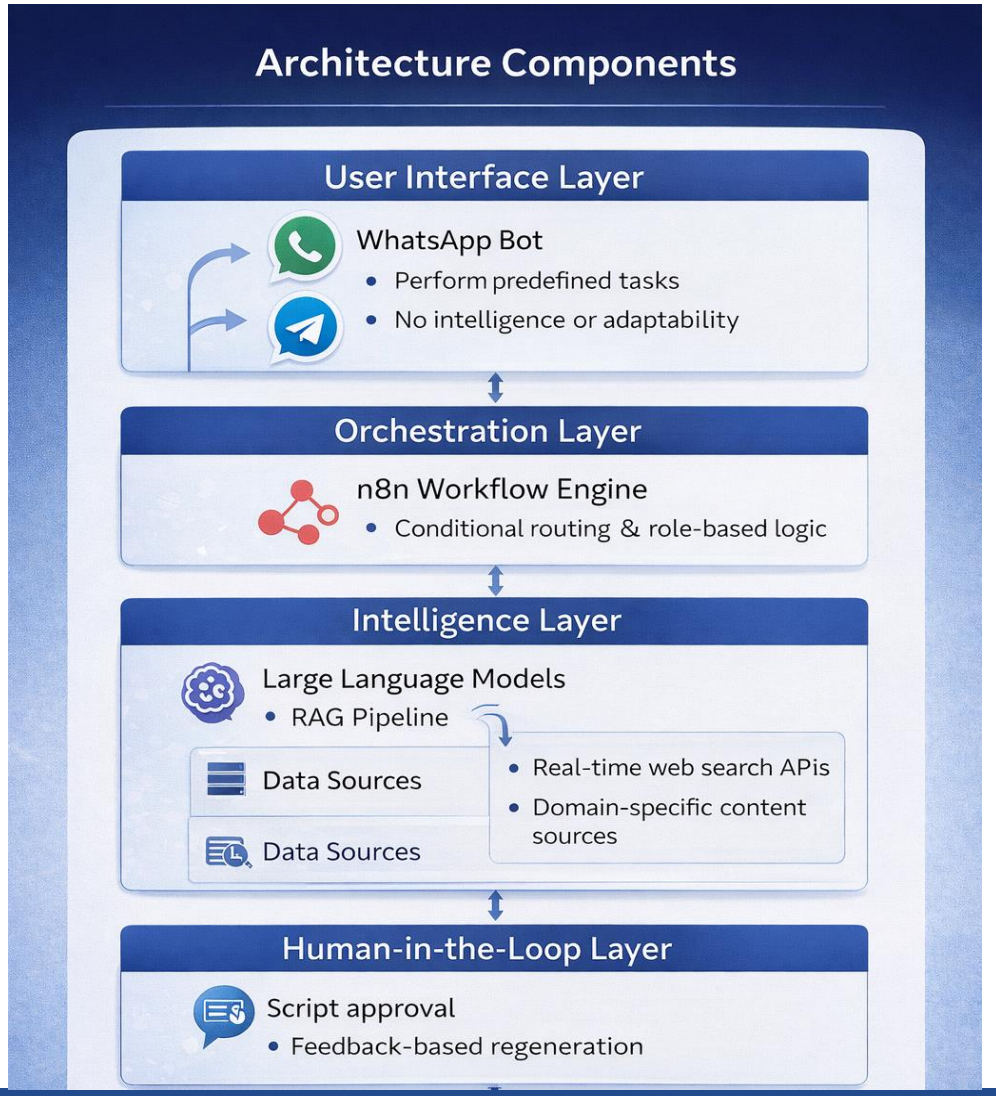
Research Gap Identified :

- Lack of end-to-end agentic systems
- No structured human-in-the-loop appproval

OBJECTIVES

- **Phase – I:**

  - **To develop the core Agentic AI pipeline** with WhatsApp and Telegram triggering, real-time retrieval, and RAG-based script generation.
  - **To implement a Human-in-the-Loop approval system** for controlled validation and iterative script refinement.

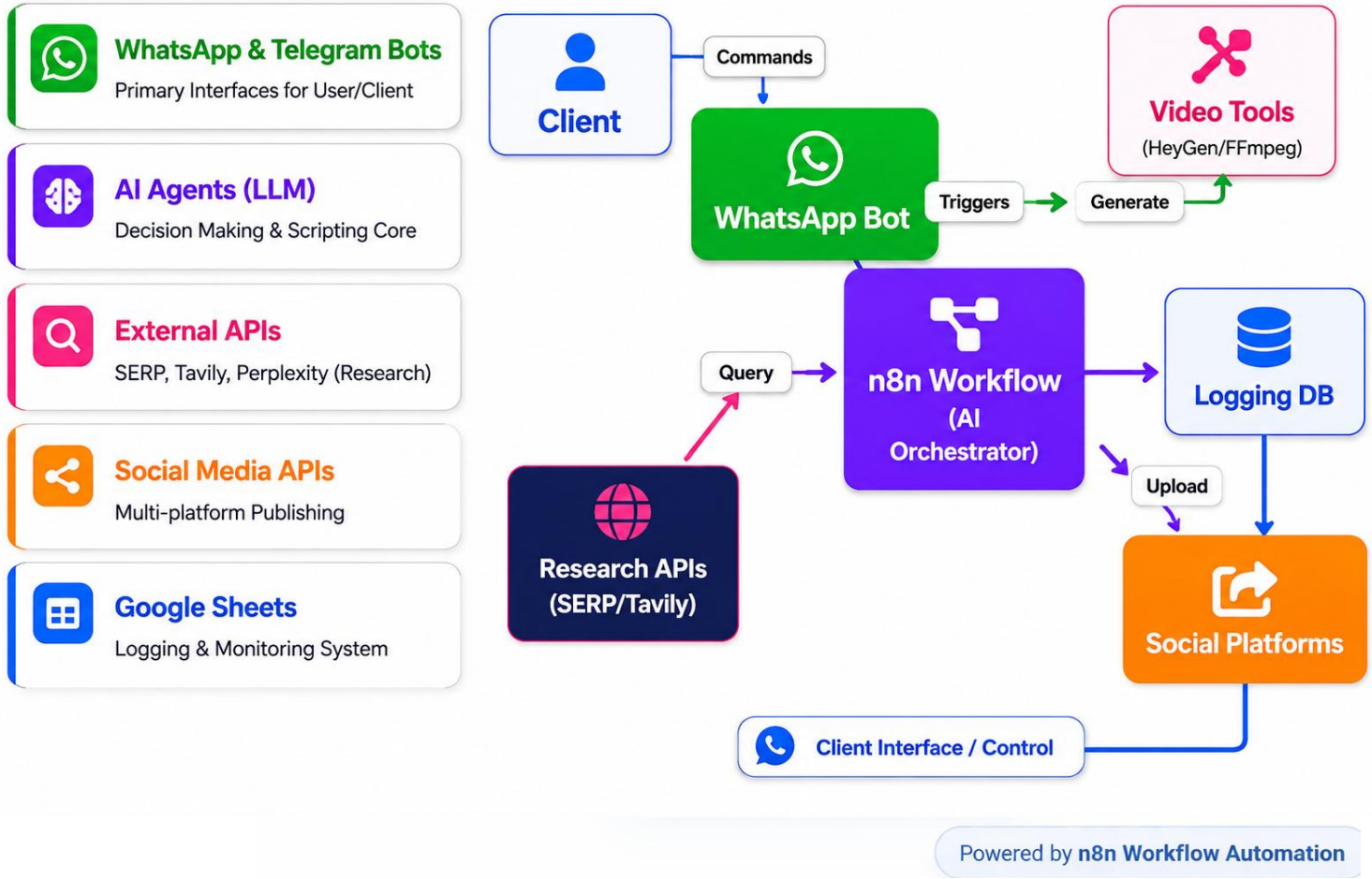


OBJECTIVES

- To design and develop a scalable Agentic AI system that delivers a **complete end-to-end content automation pipeline**.
- **Phase – II:**
  - **To automate AI-driven video generation**
    - Avatar creation
    - Voice cloning
    - Background integration
    - Editing and Merging Intro, Outro
  - **To enable multi-platform publishing and system monitoring** for scalable, production-ready deployment.
    - Publish on Youtube , Instagram, Facebook, Linkedin , etc.
    - Centralized Logging system
    - Error alert to Backend Team

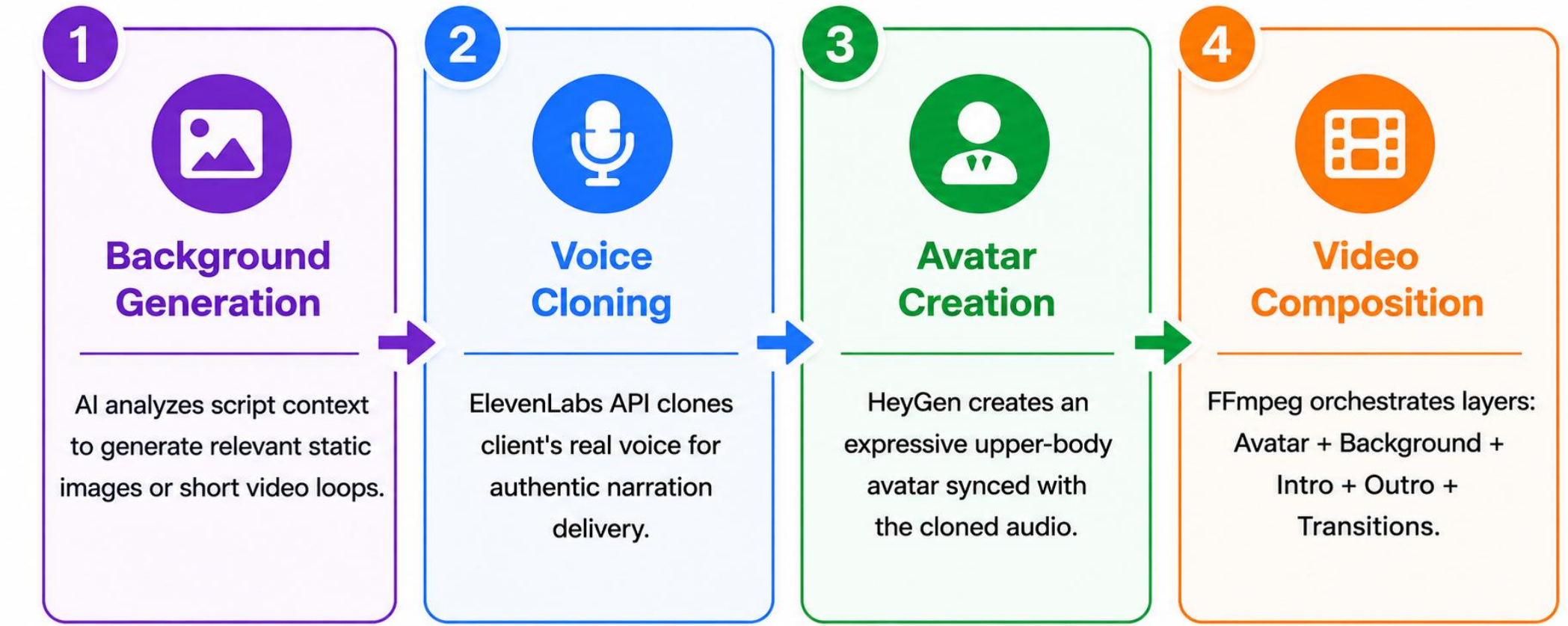
METHODOLOGY

- **Conversational Interface:** Uses WhatsApp & Telegram bots as primary user interfaces for control.
- **AI Intelligence Core:** Employs LLMs, RAG, and APIs for intelligent decision making & research.
- **Media Production:** Automates script generation, voice cloning, and avatar-based video creation.
- **Multi-Platform Publishing:**Automatically distributes content to YouTube, LinkedIn, Instagram, etc.



VIDEO GENERATION PIPELINE

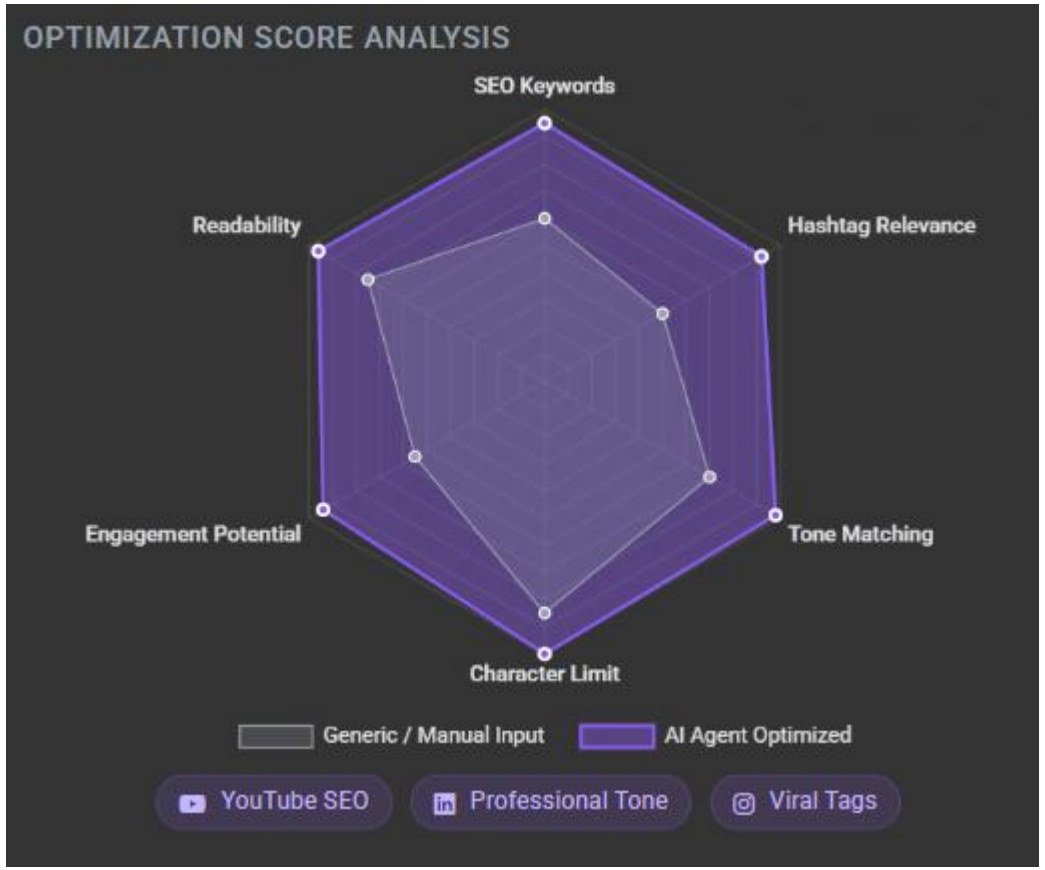
Multi-Stage AI Orchestration & Media Composition



SEO OPTIMIZED METADATA AND MULTI-PLATFORM PUBLISHING

- Once the video is approved, the AI Agent **analyzes the full script** and video context to generate metadata.

  - Generates **SEO-optimized titles designed for high click-through rates (CTR)** based on current trends.
  - Creates **Platform-Specific descriptions**
  - Intelligent **Hashtag Selection** based on video topics and platform algorithms.



LOGGING AND MONITORING SYSTEM

- **Centralized Logging :**Every workflow step is recorded in Google Sheets
- **Execution Tracking:**Logs Date, Time, and Duration for performance audits, Tokens
- **Status Monitoring:** Real-time updates on video generation & processing
- **Output Storage:** Stores final uploaded URLs for all platforms

Content_Automation_Log_2026

TIMESTAMP	CONTENT_TYPE	SCRIPT_ID	STATUS	PLATFORM_URLS
2026-01-16 09:42:15	Parsha_Weekly	PID_9283	COMPLETED	youtube.com/v/xJ...
2026-01-16 09:45:30	Legal_News	LID_4421	COMPLETED	linkedin.com/p/2...
2026-01-16 10:15:00	Legal_Update	LID_4422	GENERATING	pending...
2026-01-16 10:18:22	Short_Clip	SID_1102	QUEUED	-
2026-01-15 14:20:10	Parsha_Clip	PID_9281	COMPLETED	instagram.com/r/...
2026-01-15 14:10:05	Legal_News	LID_4419	COMPLETED	youtu.be/kL92...

ERROR HANLING & ALERT SYSTEM

Instant Notification Channels

- Immediate alerts sent to developers and clients upon system failure.

WhatsApp Telegram Email

- Ensures zero latency in error awareness for quick resolution.

Real-Time Detection

- Active monitoring of all API integrations (ElevenLabs, HeyGen, OpenAI).
- Instantly catches timeouts, authentication failures, and rate limits.
- Prevents "silent failures" where workflow stops without notice.

Rapid Debugging & Logs

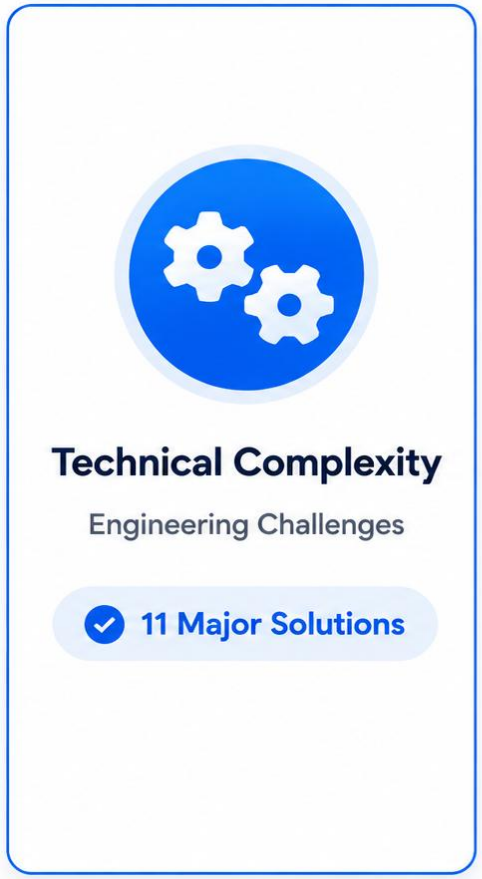
- Alerts contain direct links to the specific failed n8n execution.
- Detailed error stack traces are logged to Google Sheets.
- Enables developers to fix issues in minutes rather than hours.

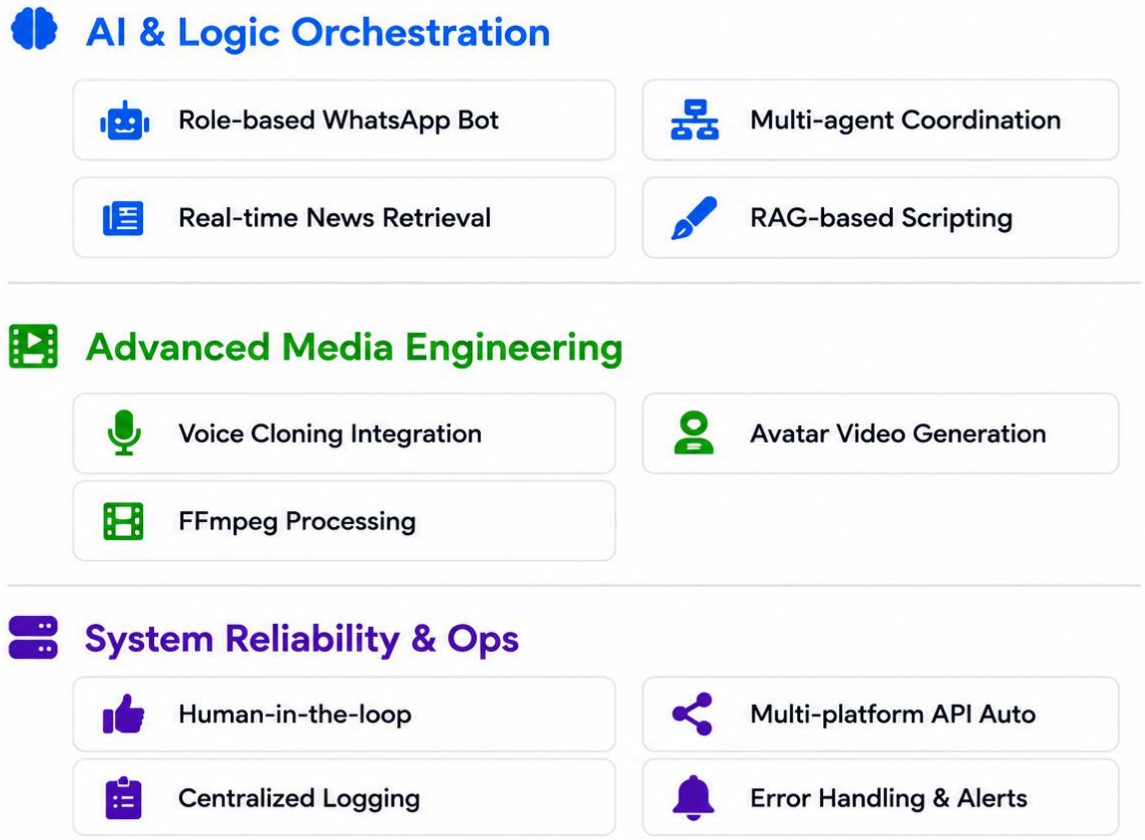
System Reliability

- Production-ready architecture designed for 24/7 uptime.
- Auto-retry mechanisms for transient network errors.
- Ensures high completion rate for complex, multi-step video workflows.

DELIVERABLES & KEY HIGHLIGHTS

- **REAL AUTOMATION:** End-to-end execution from news fetching to final video upload.
- **INDUSTRY GRADE:** Built with standard tools (n8n, FFmpeg) used in enterprise automation.
- **PRODUCTION READY:** Robust error handling, logging, and monitoring systems in place.



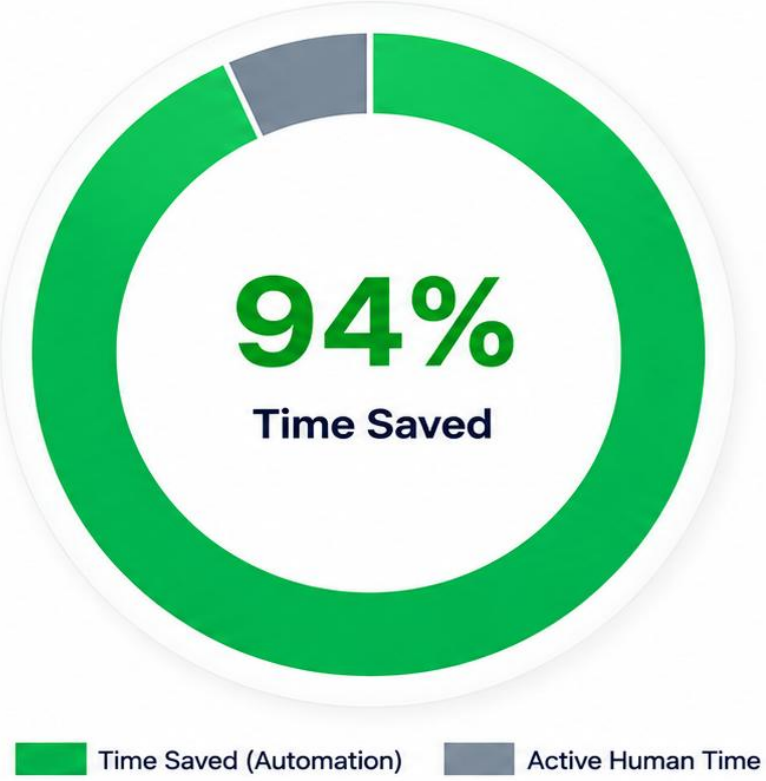


A Fully Implemented, Production-Grade System Solving Real Problems

RESULTS & IMPACT

From Hours of Manual Work to Minutes

Quantifiable Efficiency Gains & Process Improvements



Review

Drastic Time Reduction

Total production time slashed from ~4 hours to just 10-15 minutes per video cycle.

Zero Manual Effort

Eliminated manual editing, rendering, and multi-platform uploading. Human role is strictly decision-making.

Consistency & Quality

Standardized output formats and error-free metadata across all platforms, eliminating burnout.

Infinite Scalability

System architecture supports daily or even hourly publishing without additional human resource cost.

WORKFLOW PIPELINE DONE TILL NOW:

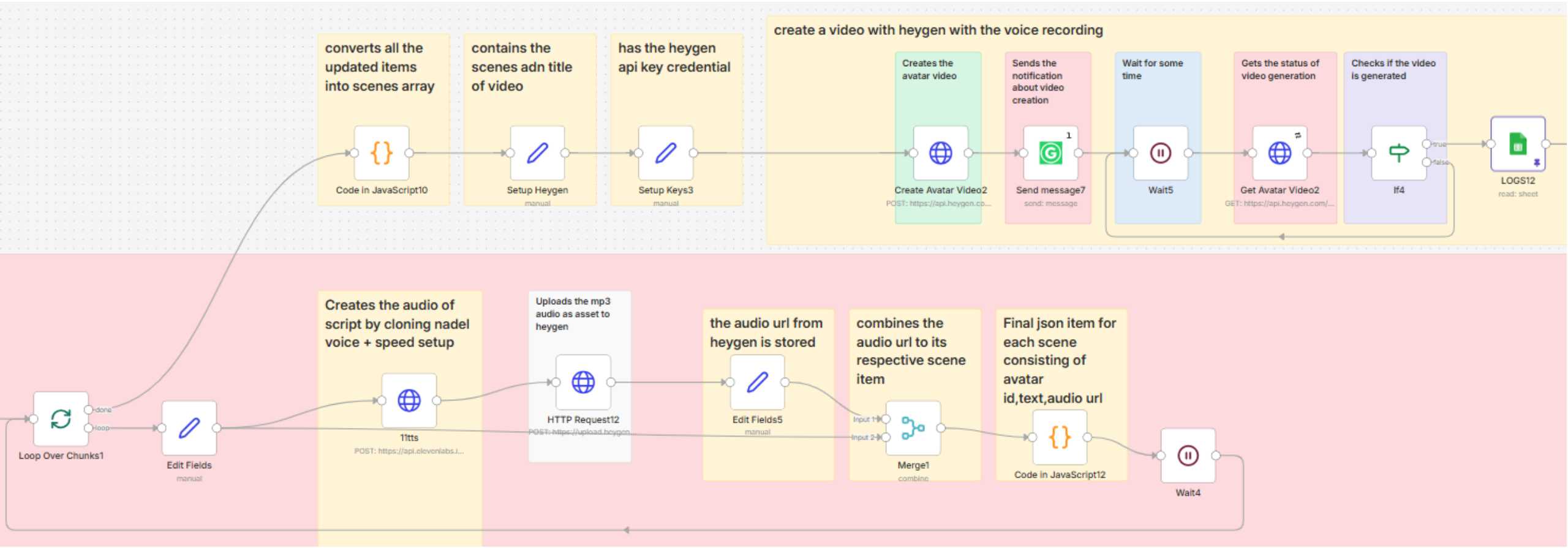


Fig. Video Generation (Voice clone+avatar clone)

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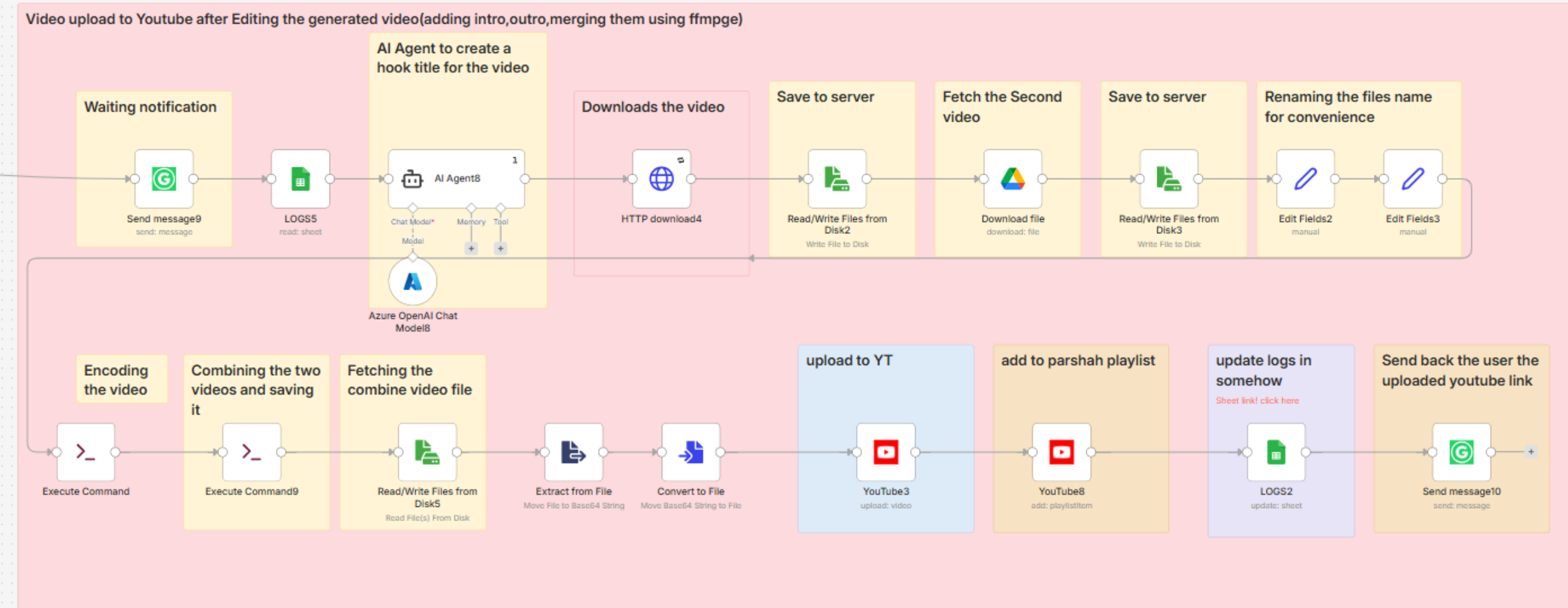


Fig. EDITING AND VIDEO PUBLISHING

FUTURE WORK PLAN & EXTENSIBILITY

- **Planned Enhancements:**
  - A Complete Web Application and Mobile Application instead of Whatsapp or telegram.
- **Cross-Domain Applicability:**
  - Marketing:
    - Instant marketing videos
  - Manufacturing:
    - CEO announcements via AI-generated video
  - Education:
    - Faculty lecture scripting
  - Corporate:
    - Policy communication
  - Healthcare:
    - Awareness content generation
- The system is **domain-agnostic** and scalable wherever time is a critical factor.

THANK YOU